

RALEIGH-DURHAM, NC, USA

WMO: 723060

Lat: 35.892N

Lon: 78.782W

Elev: 127

StdP: 99.81

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13722

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.9	-4.6	-17.5	0.8	-3.5	-14.8	1.0	-0.8	10.4	15.4	8.9	14.5	1.9	30	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	35.0	24.2	33.6	23.9	32.4	23.6	25.8	32.1	25.2	31.2	24.7	30.1	3.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.3	28.2	23.5	18.6	27.6	23.0	18.1	26.9	79.9	32.2	77.4	31.2	75.2	30.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	8.4	7.4	6.4	DB	-10.9	37.2	2.9	1.7	-13.0	38.4	-14.8	39.4	-16.4	40.3	-18.6	41.5
(5)				WB	-12.0	27.0	2.7	0.7	-14.0	27.5	-15.5	28.0	-17.1	28.4	-19.0	28.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	5.2	7.1	10.8	16.0	20.4	24.7	26.6	25.8	22.4	16.4	10.6	7.0	(6)
(7)		DBStd	8.68	5.84	5.37	5.30	4.62	3.90	2.99	2.31	2.52	3.31	4.43	4.75	5.21	(7)
(8)		HDD10.0	522	171	112	56	6	0	0	0	0	4	49	124		(8)
(9)		HDD18.3	1771	407	317	239	99	26	2	0	7	90	233	351		(9)
(10)		CDD10.0	2761	23	30	81	186	322	442	514	490	371	203	68	32	(10)
(11)		CDD18.3	969	1	1	6	28	90	194	256	232	128	31	3	1	(11)
(12)		CDH23.3	8799	1	6	51	263	777	1825	2517	2156	951	229	18	3	(12)
(13)		CDH26.7	3563	0	0	5	58	246	792	1160	925	325	51	1	0	(13)
(14)	Wind	WSAvg	2.7	2.8	3.0	3.2	3.4	2.9	2.7	2.6	2.3	2.5	2.3	2.3	2.5	(14)
(15)	Precipitation	PrecAvg	1119	87	83	102	79	96	95	120	110	107	80	80	83	(15)
(16)		PrecMax	1530	180	182	198	220	195	265	268	309	554	238	229	169	(16)
(17)		PrecMin	816	22	10	26	6	15	8	23	23	6	0	12	9	(17)
(18)		PrecStd	170	40	39	38	42	38	60	60	59	90	51	54	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.0	24.1	27.3	30.6	32.9	36.1	37.3	36.5	34.0	30.5	25.7	23.2	(19)
(20)		MCWB	16.3	16.8	17.3	19.7	22.6	24.1	24.7	24.8	22.0	21.9	18.2	17.6	(20)	
(21)		2%	DB	19.6	21.1	24.7	28.4	31.1	34.1	35.1	34.4	31.8	28.0	23.2	20.7	(21)
(22)		MCWB	15.4	14.7	16.2	18.6	21.7	23.7	24.6	24.3	22.4	20.7	16.8	16.9	(22)	
(23)		5%	DB	17.3	18.8	22.5	26.5	29.5	32.6	33.7	32.9	30.2	25.8	21.2	18.4	(23)
(24)		MCWB	14.0	13.6	15.1	17.8	21.2	23.5	24.4	23.9	22.1	19.6	16.4	15.5	(24)	
(25)		10%	DB	14.8	16.4	19.9	24.4	27.8	31.1	32.2	31.4	28.5	24.0	19.1	15.9	(25)
(26)		MCWB	11.2	11.9	13.8	17.0	20.4	22.8	24.1	23.5	21.5	18.9	15.1	12.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.4	18.4	19.3	21.7	24.4	25.8	26.7	26.6	25.2	24.2	20.6	19.5	(27)
(28)		MCDWB	20.4	21.9	24.0	27.0	29.7	32.3	33.4	33.2	30.0	27.5	22.2	22.1	(28)	
(29)		2%	WB	16.5	16.6	17.9	20.4	23.2	25.1	25.9	25.7	24.2	22.7	19.0	17.8	(29)
(30)		MCDWB	18.8	19.6	22.3	25.6	28.5	31.6	32.5	31.8	28.4	25.7	21.4	19.6	(30)	
(31)		5%	WB	14.5	14.8	16.7	19.2	22.3	24.4	25.3	25.1	23.5	21.2	17.5	15.9	(31)
(32)		MCDWB	16.9	18.0	20.7	24.3	27.3	30.5	31.6	30.7	27.2	24.4	20.1	18.2	(32)	
(33)		10%	WB	11.7	12.2	15.2	18.1	21.4	23.8	24.7	24.5	22.8	19.9	15.6	13.3	(33)
(34)		MCDWB	14.1	15.3	19.0	22.7	26.1	29.3	30.5	29.2	26.1	22.9	18.6	15.3	(34)	
(35)	Mean Daily Temperature Range		MDBR	10.4	11.2	12.0	12.4	11.6	11.1	10.6	10.3	10.2	11.7	12.0	10.5	(35)
(36)		5% DB	MCDDBR	13.1	14.0	14.5	14.4	13.0	12.8	12.5	12.2	12.2	13.2	13.9	12.5	(36)
(37)			MCWBR	9.4	9.2	7.7	6.8	5.7	4.5	4.0	3.9	4.6	6.7	8.5	8.9	(37)
(38)			MCDDBR	11.7	12.2	12.0	11.7	11.2	11.3	11.0	10.4	9.3	10.3	10.8	11.0	(38)
(39)		5% WB	MCWBR	9.6	9.3	8.0	6.5	5.3	4.5	4.1	4.0	4.3	5.9	8.0	9.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.333	0.366	0.410	0.459	0.506	0.548	0.537	0.426	0.380	0.333	0.318	(40)	
(41)		taud	2.493	2.436	2.382	2.258	2.154	2.066	1.965	1.981	2.290	2.397	2.489	2.520	(41)	
(42)		Ebn at Noon	885	906	902	875	833	791	754	753	829	840	855	858	(42)	
(43)		Edh at Noon	83	98	113	135	152	166	182	176	123	101	83	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.56	3.24	4.28	5.39	5.97	6.40	6.04	5.46	4.54	3.68	2.81	2.22	(44)	
(45)		RadStd	0.17	0.38	0.30	0.40	0.46	0.44	0.36	0.28	0.39	0.47	0.23	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+120
(47)	Regional (58 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+151	+102

Nomenclature: See separate page